

Lab 2a

Data basics and summary statistics

Team members:

Today's goal: Practice describing a dataset, distinguishing variable types, and choosing and computing useful summaries without using R.

Question 1: parts of a dataset

Suppose 5 students respond to a survey about campus events, and the data is recorded into an object named `student_survey`. Below is the same data presented in two formats: how the data can be presented to a human and how we could store the data on the computer using R.

Student	Class	Commute (mins)	Events attended	Preferred event	Plans to return
101	First year	10	1	Career panel	Yes
102	Junior	35	3	Workshop	No
103	Sophomore	0	0	Workshop	Yes
104	Senior	50	4	Career panel	No
105	First year	15	2	Social	Yes

```
student_survey
```

```

  student      class commute events_attended  preference plans_to_return
1     101 First year      10                1 Career panel         Yes
2     102   Junior      35                3   Workshop          No
3     103 Sophomore       0                0   Workshop          Yes
4     104   Senior      50                4 Career panel         No
5     105 First year      15                2    Social          Yes

```

Question 1a

What is an observation in this dataset? Who or what does each observation describe?

Question 1b

How many observations and how many variables are shown?

Question 2

Classify each variable by its variable type and ideal R storage type. Be as specific as possible: numerical discrete, numerical continuous, categorical nominal, or categorical ordinal.

variable	statistical variable type	ideal R storage type
student		
class		
commute		
events_attended		
preferred		
plans_to_return		

Question 3

Diego says, “`student` is numerical because it contains numbers.” Do you agree? Explain what makes a variable numerical rather than merely stored with numbers.

Question 4: Summarizing a categorical variable**Question 4a**

Complete the frequency table for `preferred_event`.

Preferred event	count	proportion	percent
Career panel			
Social			
Workshop			
Total			

Question 4b

Write one complete sentence interpreting a percentage from your table in context.

Question 5: Summarizing a numerical variable

The commute times, already ordered, are: 0, 10, 15, 35, 50.

Question 5a

Calculate each summary. Show your work as appropriate.

- Mean:
- Median:
- Range (distance between minimum and maximum values):
- 40th percentile:

Question 5b

Interpret the median commute time in context.

Question 5c

One student corrects their commute time from 50 minutes to 100 minutes. Without recalculating exact values, predict what happens to each statistic. Write **increases**, **decreases**, or **stays the same**.

- Mean:
- Median:
- Variance:
- Standard deviation:
- First Quartile
- IQR:

Which measure(s) are more resistant to this change of a single extreme value? Explain briefly.

Question 6

Match each research question to the most useful summary statistic. A statistic may be used more than once. Choices: count/proportion, mean, median, standard deviation, IQR.

- What is a typical number of events attended?
- Do most respondents plan to return to a campus event?
- How much do commute times typically vary around their mean?
- What is a typical commute time, when the data has one extremely large value?
- How spread out is the middle 50% of commute times?

Question 7

For each claim, decide whether the evidence supports it. If not, identify what is wrong.

- “The majority of surveyed students have attended a campus event.”
- “Because 60% of these respondents plan to return, 60% of all CSUCI students plan to return.”
- “The typical respondent attends 2 events.”
- “Students having a long commute makes them not plan to return to a campus event.”